

Kwirth

Kubernetes Observability & Management

Simple. Real-time. Open Source.

v0.5.40

kwirthmagnify.dev · github.com/kwirthmagnify/kwirth



*Kubernetes Wirth
in honor of Niklaus Wirth*

What for?

Observe & manage Kubernetes can be... complex, hard and expensive.

ELK Stack

Complex, pricey, heavy

DataDog / Dynatrace

Expensive. Very expensive

Prometheus + Grafana

Powerful, but too much config and storage

kubectl logs

Functional, but (very) limited

Lens

Just desktop experience, high resource consumption

K9s

Functional, but just console and very limited

What if you can find something simpler, that can be started with just a kubectl apply?

What is Kwirth?

An OSS **observability** and **management** tool for Kubernetes.

A single pod. No persistence. No PVs or PVCs.

Logs, metrics, alerts, events, AI, management... — in real time.

1 pod

for deploying it all

9

available channels

0

Storage needs

OSS

MIT License

```
kubectl apply -f https://...kwirth.yaml ← that's all
```

Kwirth = Kubernetes + Wirth

Niklaus Wirth

- Creator of Pascal, Modula, Oberon
- 1984 Turing Award
- Wirth's Law: "Software is getting slower more rapidly than hardware is becoming faster"
- Passed away on January 1, 2024

Magnify & our logo

The stylized "K" in the style of Roy Lichtenstein.

Just as he used Ben-Day dots to create art from comics, Kwirth takes the cluster's "dots" (logs, metrics, events) and magnifies them into something clear and visual.

Magnify: the management channel + the desktop app.

Channels

Every feature is a channel — pluggable, independent, extensible.

Log

Aggregated log streaming in real time

Metrics

CPU, memory, I/O, bandwidth...

Alert

Log-based alerting

Trivy

Cybersecurity 100%

Ops

Shell, restarts, cluster info

Fileman

Containers' filesystems access

Magnify

Full Kubernetes management platform

Pinocchio

Kubernetes extended with LLM / AI

Echo

Reference channel for channel devs

Architecture

BACKEND (Node.js / TypeScript)

Kwirth Core (WebSockets, seguridad)

Back Channels (Log, Metrics, Alert...)

Providers

Storage in etcd (secrets + configmaps)

WS / HTTP

FRONTEND (React / TypeScript)

Core Front App (auth, store, routing)

Front Channels (plug & play)

Kwirth Desktop (Electron)

Multi-cluster

Deploy: Kubernetes · Docker · Standalone (npm) · Desktop App (Electron)

Tech Stack



Kubernetes

Target platform



TypeScript

Front + Back



React

Frontend SPA



Node.js

Backend runtime



React MUI

Component library



Docker

Containerized



Helm

K8s packaging



GitHub

Source & website

WebSockets for real-time data streaming · etcd as a storage (via Kubernetes secrets/configmaps)

How to deploy Kwirth?

Kubernetes

```
kubectl apply -f kwirth.yaml  
  
helm install kwirth kwirth/kwirth -n kwirth \\  
--create-namespace
```

Kwirth inside-cluster with direct access to Kubernetes API.

Docker

```
docker run kwirthmagnify/kwirth
```

It is the perfect solution if you want to run Kwirth out-of-cluster, using your own kubeconfig.

Iron

```
npm i -g @kwirthmagnify/kwirth-external
```

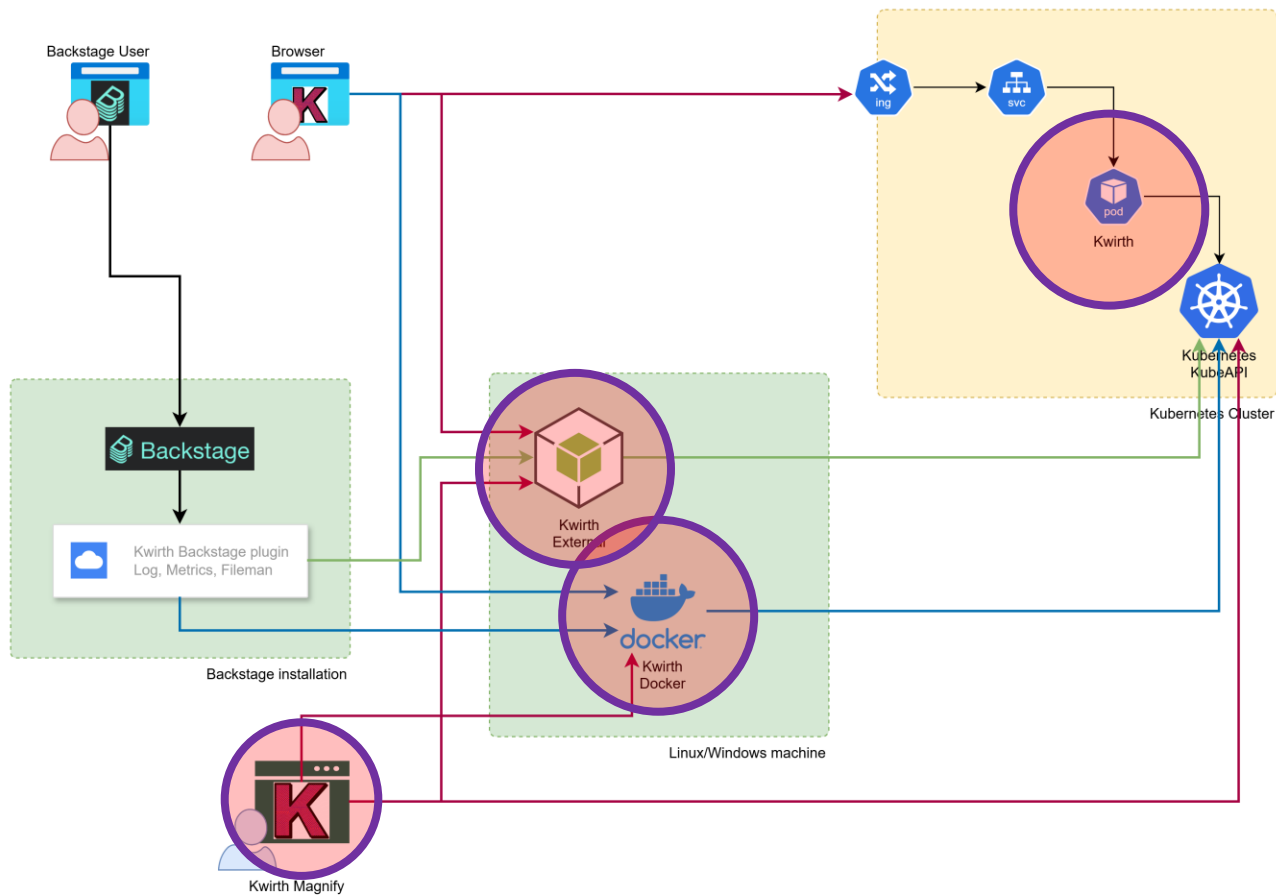
Deploy on top of your OS, next to client application (e.g., Backstage)

Desktop (Electron)

Download from kwirthmagnify.dev

Windows & Linux. Like Lens/K9s, but with the power of Kwirth.

Kwirth family



Deployment

Same codebase, four different deployment options

Total interoperability & interchangeability

Front/back fully decoupled. 100% APIfied.

[DEMO]

Let's see Kwirth in action 

1. Real time log streaming
2. Artifact metrics & aggregation
3. More fun stuff with Fileman & Trivy
4. Magnify channel — cluster management
5. Pinocchio — ask your LLM about your cluster

Roadmap

Ready

✓ Helm chart deployment

✓ Real time aggregable metrics

✓ Event streaming

✓ WebSocket multi-instance

✓ 9 channels (observability & management)

✓ Log ephemeral (alerting without storage)



In the pipeline

→ Alerts based on metrics (thresholds)

→ Pinocchio (more AI)

→ Session management, SSO, IdP, IAM

→ External notifiers

→ OTel provider, IaC support (ArgoCD, flux, kustomize...)

→ Plugin system like Backstage or Grafana

What can you do?

Kwirth has been designed for receiving everyone's help, adding the value that comes from each one's effort

Frontend

- New React MUI channels
- UX/UI improvements
- Desktop app (Electron, Tauri) and Mac
- Mobile version

Backend

- Implement more back-channels
- OpenShift support
- WebSocket features
- New providers

QA / Testers

- E2E tests
- Channel testing
- Security coverage
- Trivy integration testing

Architecture

- Design a plugin system
- Performance & scaling
- Multi-cluster strategy
- Channel spec evolution

How to join the project?

01

Clone the repo

github.com/kwirthmagnify/kwirth
Analyze the README and the docs on the website

02

Create your own environment

Node.js v22+, kubectl, Docker.
Follow the developing guide for building a new channel.

03

Select where to work

Open issues in GitHub if needed. Comment about everything you like and dislike, propose a new idea. All profiles are welcomed.

04

Pull Request

Fork & open a PR. We review it quickly,
Project is small and you'll get direct feedback.

Why to join?



Real tech

TypeScript, React, Node.js, Kubernetes.
Everything you already know and use.



Work deeply with K8s

Working with Kworth will lead you to
learn a lot of Kubernetes.



OSS in your CV

This is an open project with real
people.
It is more than a private side project.



Small community = great impact

Working in big projects gives you low
visibility.
Here, your PR changes something real.



Modern architecture

Plugin system, channels,
real-time streaming, AI. No legacy.



Real needs

Kubernetes observability and
management is a problem everyone
needs to face.

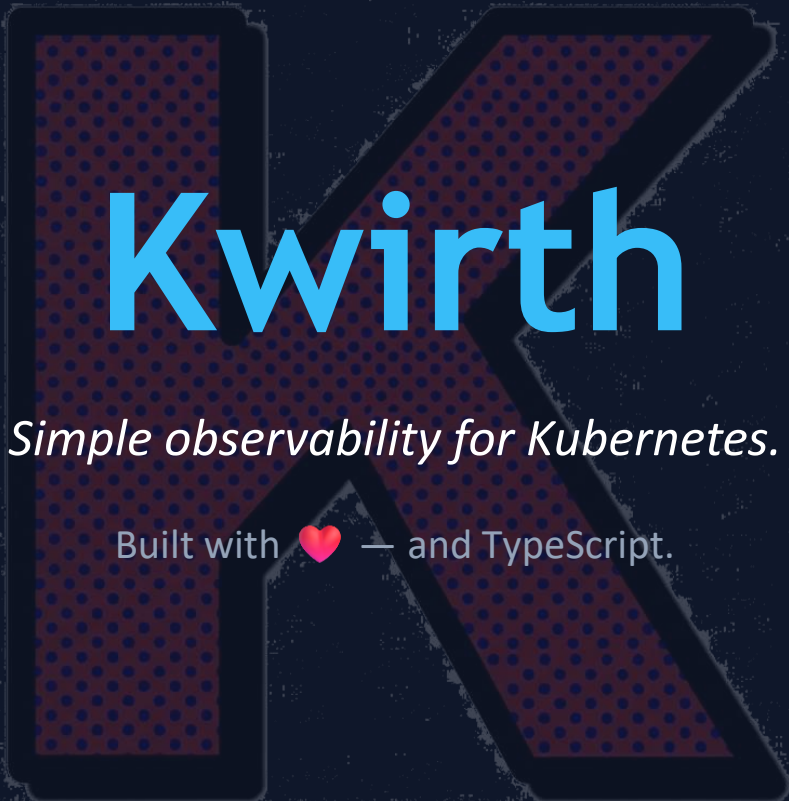
Questions?

Documentation: kwirthmagnify.dev

Source code: github.com/kwirthmagnify/kwirth

Install: `kubectl apply -f ...kwirth.yaml`

Join — Kwirth will be build by a strong community! 🙌



Kwirth

Simple observability for Kubernetes.

Built with ❤️ — and TypeScript.

kwirthmagnify.dev